Bingqian Chen

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SUMMARY

Accelerated B.S./M.S. student at Texas A&M University (B.S. Geographic Information Science & Technology, M.S. Geography). Hands-on experience in hot spot analysis, land cover classification, and web GIS applications, from field data collection to finished cartographic products. Proficient in ArcGIS Pro, Python, and SQL. Seeking a GIS Analyst / Geospatial Data Intern role to leverage spatial modeling, Python automation, and cartographic design.

EDUCATION

Texas A&M University College Station, TX
Master of Science in Geography Expected May 2028
Bachelor of Science in Geographic Information Science and Technology Expected Dec 2026

RESEARCH EXPERIENCE

Undergraduate Research Assistant Sept 2025 – Present
City Analytics & Informatics Research Group, Texas A&M University College Station, TX

- Collaborate with the research team to visualize complex resilience research results in ArcGIS Pro, translating technical findings into accessible, interactive map-based narratives.
- Prepare and integrate spatial datasets for mapping, maintaining consistent projections, symbology, and data quality across project outputs.

ACADEMIC PROJECTS

Texas Coast Hurricane Risk Analysis · *ArcGIS Pro, ArcGIS Online* Sept 2025 – Nov 2025

- Automated the batch ingestion, reprojection, and clipping of raw NOAA storm tracks and HIFLD infrastructure layers using an ArcPy Python script, standardizing datasets across Texas coastal counties.
- Applied kernel density and hot spot analysis to historical hurricane tracks to map areas of concentrated hurricane exposure along the Texas coast.
- Overlaid exposure results with infrastructure locations using intersect and spatial join tools to identify high-risk evacuation zones and the critical facilities within them.
- Developed an interactive ArcGIS Online web application with layered hazard and infrastructure data and configured pop-ups, improving access to disaster response information for emergency planning.

Aggie Park Green Space Analysis · *ArcGIS Field Maps, ArcGIS Online, GNSS* Mar 2025 – May 2025

- Mapped Aggie Park green spaces and early signs of turf degradation in the field with GNSS receivers and ArcGIS Field Maps as part of a six-person team.
- Built an ArcGIS Online field map of green-space polygons and degradation points to locate the most affected areas.
- Compared field results with past and current satellite imagery and with historical (1980–2009) and projected (2010–2039, RCP 8.5) plant hardiness zones.
- Identified degrading grass fields at risk of soil erosion and recommended irrigation, soil management, and resilient plant species in a research poster.

U.S. Economic Disparities Analysis · *ArcGIS Pro, ArcGIS StoryMaps* Sept 2024 – Nov 2024

- Retrieved U.S. Census Bureau ACS 5-year tabular estimates and executed SQL queries to filter and join household income and educational attainment attributes to county geometries in ArcGIS Pro.
- Designed a bivariate thematic map with a combined classification and color scheme to reveal where low household income and low educational attainment coincide.
- Developed an interactive ArcGIS StoryMap integrating maps, spatial analysis, and regional case studies to communicate patterns of economic inequality to a general audience.

TECHNICAL SKILLS

GIS & Mapping: ArcGIS Pro, ArcGIS Online, ArcGIS Field Maps, QGIS, Google Earth Engine

Spatial Methods: Supervised / Unsupervised Land Cover Classification, Overlay Analysis, Kernel Density Estimation, Hot Spot Analysis and Bivariate Mapping

Programming & Data: Python (ArcPy, geoprocessing automation, pandas), SQL (attribute queries, table joins), R (statistical analysis)

Languages: Mandarin and Cantonese (native), English (professional working proficiency)